

TBA4245 GEODESY, spring 2018. Assignment no. 2:
The Geodesic (Geodetisk linje)
Solution

NB: See hints on page 2 in the assignment text. In this solution the Holsen programs are used, also where manual calculations are required.

Latitude and longitude in WGS84 for Roma and New York:

	Latitude, B	Longitude, L	Longitude, L
Roma	41.88333333° N	12.50000000° E Gr.w.	12.50000000° E Gr.w.
New York	40.66666667° N	73.83333333° W Gr.w.	-73.83333333° E Gr.w.
Mean value	41.27500000° N		

NB: $l = 12.5 - (-73.83333333) = 86.33333333$, see the figure on the next page.

NB: It is recommended to use ca. 8 decimals on B and L , to have the cm accuracy:

$$1 \text{ Arc second in } B \text{ is ca } 30 \text{ m: } 1'' = \frac{1}{60 \cdot 60}^\circ = 0.00027778^\circ, \Rightarrow 3 \text{ cm equals } 0.00000028^\circ$$

Calculated reduced latitudes, β : $\beta_R = 41.78770742^\circ$ $\beta_{NY} = 40.57157716^\circ$

(a) Calculate manually how much further north Rome is, **in meters**?

The difference in the length of the meridians will give the answer.

Approximate formula in Holsen "Geodesy part 1", formula (2.48): 135122.059 m

Exact formula from Holsen "Geodesy part 1", formula (2.47) or (2.46):

The length of the meridian from Equator: G to Rome = 4638678,462

G to NY = 4503556,396

The difference: 135122.066 m

The Holsen program MERIDBUE: 135122.066 m

(b) The Geodesic

Calculate manually the azimuth of the geodesic in the endpoint NY, from NY to R.

Holsen: $A_{R-NY} = 300.900429295^\circ$ (given in the assignment text)

Using one of Clairaut's formulas, the 2nd, see chapter 1.6 in "Geodesy part 2":

$$\cos \beta \cdot \sin A = k = \text{constant} \quad (\text{reduced latitude, } \beta), \quad k = -0,639786615$$

The constant k has a constant value in all points on the geodesic R-NY. The k is calculated in R, and the azimuth in NY can be calculated by Clairaut's 2nd:

Clairaut: $A_{NY-R} = 237.38107519^\circ$, NB: From NY and westwards, same direction as in R)

Towards east: $A_{NY-R} = 237.38107519 + 180 = 57,38107519^\circ$

Details:

From Clairauts: $\sin A = -0.842274396$

Two solutions, the calculator gives $A = -57,381...$, the other solution is $A = 237,381...$ and this is the correct value, check the figure.

Turning the azimuth towards Rome: $A = 237,381... - 180 = +57,381...$

Calculate manually the latitude of the northernmost point of the geodesic going through R and NY.

Clairaut's formulas, the 2nd: $\cos \beta \cdot \sin A = k = \text{constant}$, see above for the value of k .

At maximum latitude the azimuth of the geodesic is 90° .

$$\beta_0 = \text{inv cos } k =$$

Converting to geographic latitude: $B = 50.31867836^\circ$

- (c) Calculate manually the length of the arc of the normal section from R to NY and from NY to R. Why will we generally obtain two different values of these two distances?

Starting point: The length of the geodesic, given in the assignment text: $S = 6899421.417$ m
The geodesic is the shortest line.

Using formula (1.7) in "Geodesy part 2", $e'^2 = 0,00673949676$ (for the WGS84-ellipsoid).

Formula of η , eta, used in (1.7b): $\eta^2 = e'^2 \cdot \cos^2 B$

$$(1.17a) \quad \Delta S = S_G - S_N = -\frac{e^4 \cdot S^5}{360 \cdot a^4} \cdot \cos^4 B \cdot \sin^2 2A + \dots$$

Formula (1.7) gives the difference in length between the geodesic (S_G) and the normal section (S_N):

$$\text{From R to NY:} \quad S_G - S_N = -0.281 \text{ m} \quad \Rightarrow S_N = 6899421.698 \text{ m}$$

$$\text{From NY to R:} \quad S_G - S_N = -0.321 \text{ m} \quad \Rightarrow S_N = 6899421.738 \text{ m}$$

The lengths will generally be different, as the traces on the ellipsoid differ, see figures 1.1 and 1.13 in "Geodesy part 2". If the points are situated on the same latitude, the lengths will be the same.

Calculate manually the difference between the azimuth of the geodesic and the azimuth of the arc of the normal section from R to NY and do similar calculations from NY to R. Comment on the results.

Same reason here: The lengths will generally be different, as the traces on the ellipsoid differ. The values will have opposite signs (plus/minus), see figure 1.13 in "Geodesy part 2" for a sketch. As Rome is the northernmost point, Rome corresponds with P2 on that figure.

Difference in Azimuths, formula (1.6a) in "Geodesy part 2":

$$(1.16a) \quad \Delta A = A_G - A_N = -\frac{e^2 \cdot S^2}{12 \cdot a^2} \cdot \cos^2 B \cdot \sin 2A + \dots$$

$$\text{In R: } \text{diff} = A_{\text{Geodesic}} - A_{\text{NormalSectionArc}} = +0.0182706^\circ$$

$$\text{In NY: } \text{diff} = A_{\text{Geodesic}} - A_{\text{NormalSectionArc}} = -0.0195401^\circ$$

For the questions (d) and (e): We simplify the calculations by saying the two points have the same value of the latitude, use the mean value of the latitude of the two points, R and NY:

Point	Latitude	Longitude
RS	41° 16' 30,0000" N	12° 30' 0,0000" East Gr.w.
NYS	41° 16' 30,0000" N	73° 50' 0,0000" West Gr.w.

- (d) Calculate manually the length of the circle of latitude (breddesirkelen) between RS and NYS in meters, when you:

- use a sphere with a radius of an average value
- use the WGS84 ellipsoid

Sphere, see page 2 in the assignment text (radius which gives the same volume as the ellipsoid)

$$R = 6371000.79 \text{ m}$$

$$\text{The radius of the circle of latitude: } R \cdot \cos B = 4788138.654 \text{ m}$$

$$\text{The difference in longitude: } 12.5 + 73.8333333 = 86.3333333^\circ$$

The length of the arc of the circle of latitude, $\text{Arc} = R \cdot \Delta L$, L in radians:

$$4788138.654 \cdot (86.3333333 \cdot \pi/180) = 7214771.732 \text{ m}$$

Spheroid/ellipsoid:

$$\text{WGS84 ellipsoid, latitude (mean value)} = 41.275^\circ$$

$$(\text{Radius of curvature in the Meridian: } M = 6363224.993 \text{ m})$$

Radius of curvature in Prime vertical for the mean latitude, using formula (2.48) in “Geodesy part 1”:

$$N = 6387447.711 \text{ m}$$

$$\text{The radius of the circle of latitude: } N \cdot \cos B = 4800499.371 \text{ m}$$

$$\text{The difference in longitude: } 12.5 + 73.8333333 = 86.3333333^\circ$$

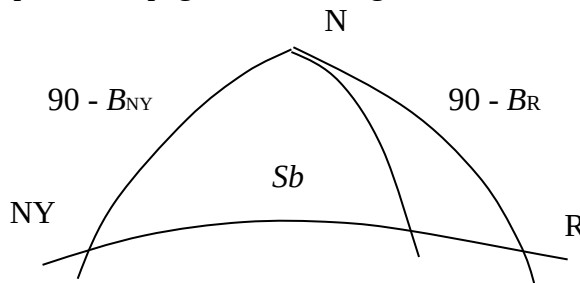
The length of the arc of the circle of latitude, $\text{Arc} = R \cdot \Delta L$, L in radians::

$$4800499.371 \cdot (86.3333333^\circ \cdot \pi/180^\circ) = 7233396.872 \text{ m}$$

$$\text{If the reduced latitude is used instead of geographic latitude: } 7225303.748 \text{ m}$$

- (e) *On the sphere. Great circle and normal section arc:
Calculate manually the length (in meters) of the great circle arc between RS and NYS, if you assume a sphere with an average radius (which radius?).
Comment on the results in (d) and (e).*

We choose to use a sphere with an average radius: $R = 6371000.7900 \text{ m}$, same volume as the WGS84 ellipsoid, see page 2 of the assignment text.



On the sphere:

We have to find the angle in N between R and NY. By using spherical trigonometry (see chapter 3 in “Geodesy part 1”). In the triangle N-NY-R the length (in degrees) of the arc of the great circle (Sb), by using the spherical cosine sentence for the sides of the triangle:

$$\cos Sb = \cos \Delta L \cdot \sin(90 - B_{NY}) \cdot \sin(90 - B_R) + \cos(90 - B_{NY}) \cdot \cos(90 - B_R)$$

$$\cos Sb = \cos \Delta L \cdot \cos B_{NY} \cdot \cos B_R + \sin B_{NY} \cdot \sin B_R$$

Here the two latitudes of RS and NYS are equal, 41.275° :

$$\cos Sb = \cos 86.33333333 \cdot \cos^2 41.27500000 + \sin^2 41.27500000$$

The length (in degrees) of the arc of the great circle (Sb): 61.88184214°

With $R = 6371000.7900$ we obtain: $Sb = 6880947.751 \text{ m}$, when the geographic latitude B is used.

Comparing (d) and (e) we see that the great circle arc is shorter than the length of the arc of the circle of latitude. This is Ok as the great circle is a geodesic on the sphere.

The values cannot be compared directly as the values are influenced by the choice of the latitude used in the formulas: Which latitude of the reduced (β) and the geodetic (B) shall we use?